

UTM
UNIVERSITI TEKNOLOGI MALAYSIA

School of
Professional and
Continuing
Education
(SPACE)

ASSIGNMENT 2

FSPT 0024

FUNDAMENTALS OF PROGRAMMING

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Section: 24

1. Suppose a, b and c are **INTEGER** variables assigned the values a = 8, b = 3 and c = -5. Determine the value of each of the following arithmetic expressions.

   a) 2 * b + 3 * (a-c)

   b) a * b / c

   c) a * (b / c)

   d) (a * c) % b

   e) a * (c % b)

   f) b * (a + c) * b / 2

   g) 15 + c / 6 - 2 * 6 % 5

   h) ((15 + c) / (6 – 2) * 6) % 5

[Programming] Assignment 2 coding.cpp

```cpp
#include <iostream>
using namespace std;

int main()
{
    int a = 8,
        b = 3,
        c = -5;

        cout << "a: " << " 2 * b + 3 * ( a - c ) = " << 2 * b + 3 * ( a - c ) << endl;

        cout << "b: " << " a * b / c = " << a * b / c << endl;

        cout << "c: " << " a * ( b / c ) = " << a * ( b / c ) << endl;

        cout << "d: " << " ( a * c ) % b = " << ( a * c ) % b << endl;

        cout << "e: " << " a * ( c % b ) = " << a * ( c % b ) << endl;

        cout << "f: " << " b * ( a + c ) * b / 2 = " <<  b * ( a + c ) * b / 2 << endl;

        cout << "g: " << " 15 + c / 6 -2 * 6 % 5 = " << 15 + c / 6 -2 * 6 % 5 << endl;

        cout << "h: " << " (( 15 + c ) / ( 6 -2 ) * 6) % 5 = " << (( 15 + c ) / ( 6 -2 ) * 6) % 5 << endl;

        return 0;
}
```

```
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a:  2 * b + 3 * ( a - c ) = 45
b:  a * b / c = -4
c:  a * ( b / c ) = 0
d:  ( a * c ) % b = -1
e:  a * ( c % b ) = -16
f:  b * ( a + c ) * b / 2 = 13
g:  15 + c / 6 -2 * 6 % 5 = 13
h:  (( 15 + c ) / ( 6 -2 ) * 6) % 5 = 2

--------------------------------
Process exited after 0.1552 seconds with return value 0
Press any key to continue . . .
```

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2. Declare data1, data2, data3, and data4 as integer. Determine the current variable value below if data1=3, data2 =6, data3 =10 and data4=15. Please note that each expression is independent.

   a) data1 += 5

   b) data1 = data1 + 2

   c) data2 += data1

   d) data2 = data2 + data1--

   e) data3 -= 2 * --data2

   f) data3 *= data3 - 1

   g) data4 -= data3++

   h) data4 = data4 * ++data3

[*] [Programming] Assignment 2 coding.cpp

```cpp
#include <iostream>
using namespace std;

int main()
{
    int data1, data2, data3, data4;

    {   data1 = 3;
        data1 += 5;

        cout << "a) " << " data1 += 5 : " <<  data1 << endl; }


    {   data1 = 3;
        data1 = data1 + 2;

        cout << "b) " << " data1 = data1 + 2 : " << data1 << endl; }

    {   data1 = 3;
        data2 = 6;
        data2 += data1;

        cout << "c) " << " data2 += data1 : " << data2 << endl; }

    {   data1 = 3;
        data2 = 6;
        data2 = data2 + data1--;

        cout << "d) " << " data2 = data2 + data1-- : " << data2 << endl;    }
```

[*] [Programming] Assignment 2 coding.cpp

```cpp

    {   data2 = 6;
        data3 = 10;
        data3 -= 2 * --data2;

        cout << "e) " << " data3 -= 2 * --data2 : " << data3 << endl;   }

    {   data3 = 10;
        data3 *= data3 -1;

        cout << "f) " << " data3 *= data3 -1 : " << data3 << endl;  }

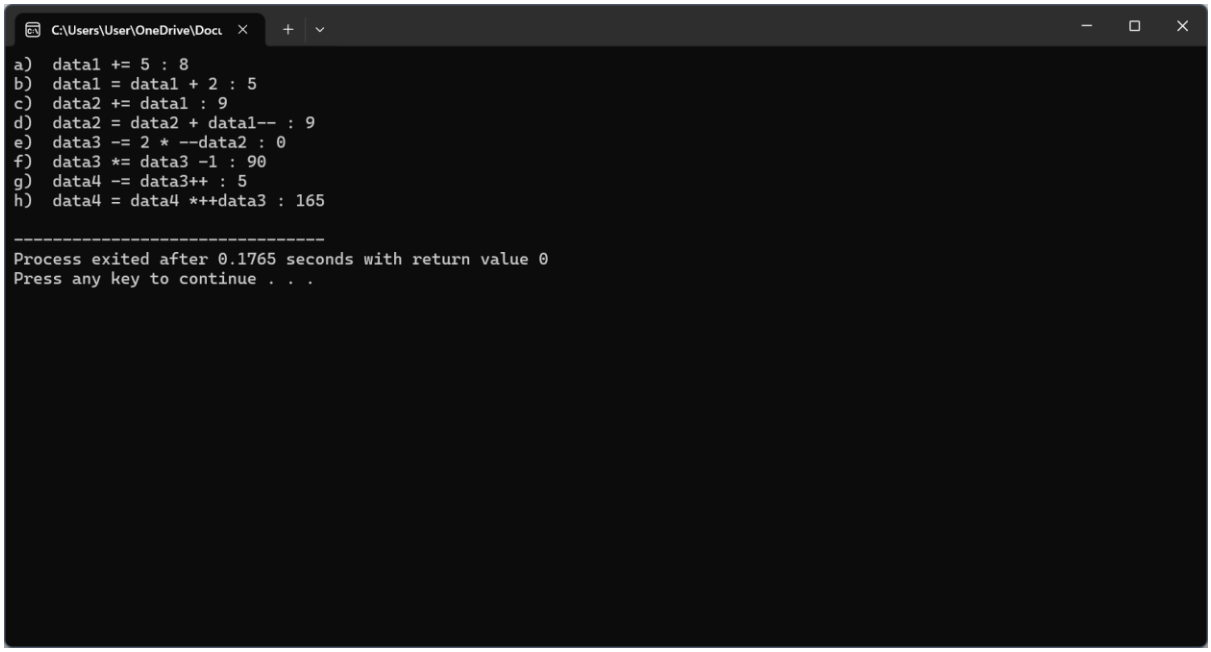
    {   data3 = 10;
        data4 = 15;
        data4 -= data3++;

        cout << "g) " << " data4 -= data3++ : " << data4 << endl;   }

    {   data3 = 10;
        data4 = 15;
        data4 = data4 *++data3;

        cout << "h) " << " data4 = data4 *++data3 : " << data4 << endl; }

        return 0;
}
```


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a) data1 += 5 : 8
b) data1 = data1 + 2 : 5
c) data2 += data1 : 9
d) data2 = data2 + data1-- : 9
e) data3 -= 2 * --data2 : 0
f) data3 *= data3 -1 : 90
g) data4 -= data3++ : 5
h) data4 = data4 *++data3 : 165

Process exited after 0.1765 seconds with return value 0
Press any key to continue . . .

3. Determine the value of the expression below.

```
int x = 3, y = 5;
char code = 'A', code2 = 'S';
bool p = false;
bool q;
cout << ((x + 3) > (y + 5)) << endl;
cout << (((p != 0) + 10) && ((p + 10) == 10)) << endl;
cout << ((code == 'C') || (code2 != 'S')) << endl;
q = x - y;
x = q + 1;
cout << "The value of q is " << q << endl;
cout << "The value of x is " << x << endl;
```

[*] [Programming] Assignment 2 coding.cpp

```cpp
#include <iostream>
using namespace std;

int main()
{
    int x = 3, y = 5;
    char code = 'A', code2 = 'S';
    bool p = false;
    bool q;

    cout << ( (x + 3) > (y + 5) ) << endl;
    cout << ( ( (p != 0) + 10 ) && ( (p + 10) == 10) ) << endl;
    cout << ( (code == 'C') || (code2 != 'S') ) << endl;

    q = x - y;
    x = q + 1;
    cout << "The value of q is " << q << endl;
    cout << "The value of x is " << x << endl;

    return 0;

}
```

```
C:\Users\User\OneDrive\Docu
0
1
0
The value of q is 1
The value of x is 2

--------------------------------
Process exited after 0.2057 seconds with return value 0
Press any key to continue . . .
```

4. The program provided contains several syntax errors. Let's identify them and rewrite the correct version:

```
*/ what is wrong with this program? /*
#include iostream
Using namespace std
int main()
{
int x, y, sum;
x = 5 y = 10;
sum = x + y
cout >> "The sum of x and y is: " << sum;
return 0;
{
```

[Programming] Assignment 2 coding.cpp

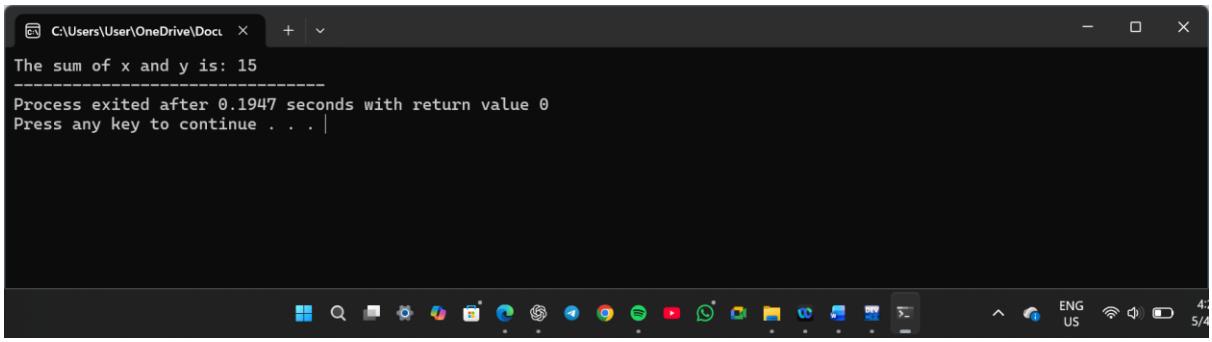
```cpp
#include <iostream>
using namespace std;

int main()
{
    int x, y, sum;
    x = 5, y = 10;
    sum = x + y;
    cout << "The sum of x and y is: " << sum;

    return 0;

}

```



5. Write a simple C++ program that prompts the user to enter two integers, calculates their product, and then displays the result as the expected output below. Refer to the explanation for your reference.

```
Enter the first integer: 25
Enter the first integer: 15
The product of 25 and 15 is 375.
```

- **Include the iostream library**: This library is used for input and output operations.
- **Using namespace std**: This allows us to use names from the `std` namespace directly (like `cout` and `cin`), which simplifies the code.
- **Declare variables**:
  - `int num1, num2, product;` declares three integer variables to store the user's input and the result.
- **Prompt the user for input**:
  - `cout << "Enter the first integer: ";` prompts the user.
  - `cin >> num1;` reads the user's input and stores it in `num1`.
  - Similarly, the second integer is read and stored in `num2`.
- **Calculate the product**:
  - `product = num1 * num2;` performs the multiplication of the two integers.
- **Display the result**:
  - `cout << "The product of " << num1 << " and " << num2 << " is " << product << ".";` prints the result to the console.
- **Return 0**: This indicates that the program executed successfully.

[Programming] Assignment 2 coding.cpp

```cpp
#include <iostream>
using namespace std;

int main()
{
    int num1, num2, product;

    cout << "Enter the first integer: ";
    cin >> num1;

    cout << "Enter the first integer: ";
    cin >> num2;

    product = num1 * num2;

    cout << "The product of " << num1 << " and " << num2 << " is " << product << ".";

    return 0;

}
```

```
C:\Users\User\OneDrive\Docu
Enter the first integer: 25
Enter the first integer: 15
The product of 25 and 15 is 375.
--------------------------------
Process exited after 11.57 seconds with return value 0
Press any key to continue . . .
```